

Application Strength / Dataset Readiness Gate

Decision Report Before Surrogate Dataset Planning

Codex-assisted downstream gate report

June 17, 2026

Abstract

This report decides whether the project has enough application-strength evidence to move from small stress analytics into surrogate dataset planning. The decision is **READY_FOR_DATASET_PLANNING_WITH_LIMITATIONS**. This is deliberately not approval for dataset construction, neural-surrogate training, machine-learning label export, broad parameter sweeps, or production risk claims. The supported next step is a written surrogate dataset design plan that preserves solver, boundary, Greek, and diagnostic cautions.

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1 Purpose of the Application-Strength Gate

The solver-validation ladder ended with Ticket 12's conservative decision `PASS_SOLVER_VALIDATION_WITH_LIMITATIONS`. Pilot 01 then showed that the validated baseline American Crank–Nicolson / projected SOR solver can produce small diagnostic stress-map figures with metadata. The quantitative risk-analytics step added numbers for boundary movement, continuation premium, Gamma concentration, PSOR effort, runtime, and LCP diagnostics.

This gate answers a narrower downstream question: are the application messages now clear enough to justify a surrogate dataset design plan? It does not create a dataset. It does not create labels. It does not train a neural surrogate. It is a human-review decision point between diagnostic analytics and any later data-generation work.

The methodology is consistent with the project planning documents and the student methodology handouts: first validate the classical free-boundary benchmark, then run risk analytics, and only later consider supervised surrogate data and models (Black and Scholes, 1973; Merton, 1973; Brennan and Schwartz, 1977; Wilmott et al., 1995).

2 Evidence Reviewed

The gate uses four evidence layers:

- Ticket 12 solver-validation synthesis.
- Downstream design gate policy.
- Pilot 01 stress-map diagnostics.
- Quantitative risk-analytics tables and figures.

Evidence source	Main evidence	Gate reading
Ticket 12 synthesis	<code>PASS_SOLVER_VALIDATION_WITH_LIMITATIONS</code> .	Baseline solver is credible for human-reviewed downstream planning, not automatic data generation.
Downstream design gate	Baseline CN/PSOR selected; Rannacher kept as comparison only.	Future design must not silently switch solvers.
Pilot 01	11 pilot runs; all <code>ACCEPTABLE_FOR_PILOT_REVIEW</code> .	Small stress-map diagnostics are feasible with required metadata.
Quantitative analytics	31 controlled analytics rows; all <code>ACCEPTABLE_FOR_ANALYTICS_REVIEW</code> .	Application messages are now quantitative enough for design planning.
Methodology handouts	Risk analytics should precede dataset and surrogate work.	The next permitted step is dataset planning, not construction.

3 Supported Application Messages

The strongest application messages are directional and diagnostic. They are not universal theorems over all parameter regimes, but they are coherent with the methodology handouts and the project’s American-option intuition.

Supported message	Evidence	Interpretation
American put boundary decreases as volatility rises.	At $\tau/T = 1$, boundary moneyness moves $0.7668 \rightarrow 0.5667 \rightarrow 0.4000$ as $\sigma = 0.20, 0.40, 0.60$.	Higher volatility increases waiting value for the put in the tested regime.
Dividend-call boundary decreases as dividend yield rises.	At $\tau/T = 1$, boundary moneyness moves $1.8979 \rightarrow 1.2666 \rightarrow 1.2332 \rightarrow 1.1666$ as $q = 0.03, 0.08, 0.10, 0.14$.	Higher dividend yield makes early exercise more attractive in the tested call regime.
Higher volatility increases numerical effort.	Put mean PSOR iterations rise from 6.99 to 29.42; call mean iterations rise from 7.35 to 14.92.	Stress regimes can be more expensive and need iteration metadata.
LCP diagnostics remain small in tested analytics cases.	Max obstacle violation is 0; max equation violation is about 4.89×10^{-8} ; max complementarity product is about 4.73×10^{-10} .	The diagnostic analytics are internally coherent for human review.
Gamma remains fragile and mask-dependent.	Full-grid max $ \Gamma = 30$; strict-mask values are lower but still diagnostic.	Greek outputs require masks and should not become production labels yet.

4 Cautionary Findings

The application evidence is strong enough for planning, but not strong enough for data generation. The caution is especially important for boundaries and Greeks. The boundary curves are extracted from the continuation premium $U - \Phi$ using a threshold. They are useful diagnostics, not exact analytical free-boundary solutions. Gamma is a finite-difference second derivative, so it is especially sensitive near payoff kinks, maturity rows, and extracted boundary bands.

Caution	Why it matters	Planning implication
Threshold boundary extraction	Boundary location depends on grid, threshold, interpolation, and time row.	Store threshold and found/not-found metadata.
Gamma fragility	Full-grid Gamma is dominated by kink and maturity behavior.	Use masks; do not export unmasked Gamma labels.
Representative cases only	The analytics use controlled one-at-a-time sweeps, not full regime coverage.	Dataset design must plan held-out regimes carefully.

Caution	Why it matters	Planning implication
Local runtime numbers	Runtime seconds are machine-specific.	Use runtime as planning evidence, not production benchmark.
Baseline solver retained	Rannacher remains comparison-only after Ticket 10A.	Label source must record solver variant explicitly.

5 Dataset-Readiness Decision

Decision: READY_FOR_DATASET_PLANNING_WITH_LIMITATIONS

Allowed next step: write a surrogate dataset design plan.

Not allowed yet: generate datasets, labels, neural models, or broad parameter sweeps.

Decision field	Value
Gate decision	READY_FOR_DATASET_PLANNING_WITH_LIMITATIONS.
Basis	Solver validation passed with limitations; pilot and analytics evidence show clear diagnostic messages.
Allowed next step	Surrogate dataset design plan only.
Main limitation	Application evidence is representative and diagnostic, not broad label readiness.
Blocked until later	Dataset construction, neural training, production Greek labels, broad sweeps, final claims across all regimes.

6 Allowed Next Step

The next step may be a written surrogate dataset design plan. That plan should define the future data schema, parameter ranges, metadata requirements, split strategy, diagnostic gates, and human review criteria. It should still avoid generating arrays or labels until a separate construction gate approves the design.

The plan should explicitly state that all future labels must come from the validated baseline CN/PSOR solver unless a later method gate approves an alternative. If any Rannacher-smoothed results are included later, they must be separately labelled and must not replace the baseline by default (Rannacher, 1984).

7 Blocked Actions

Blocked action	Reason
Actual dataset construction	This gate approves planning only; no arrays or training tables should be exported.

Blocked action	Reason
Neural surrogate training	The project has not yet approved data schema, label policy, or held-out regime design.
Training-label export	Boundary and Greek quantities remain diagnostic and metadata-dependent.
Production Greek labels	Gamma remains fragile near kink, maturity, and boundary regions.
Broad regime sweeps	Current evidence comes from controlled pilot and analytics sweeps.
Final claims across all regimes	The tested cases are representative, not universal convergence proof.
Exact free-boundary claims	Boundaries are threshold-based diagnostics.

8 Preliminary Dataset Design Rules

The following rules are not a dataset schema yet. They are constraints that the future dataset design plan must respect.

Rule area	Required planning rule	Evidence basis
Normalization	Use normalized strike $K = 1$ for the first design.	Pilot 01 and quantitative analytics.
Option families	Include American puts and dividend-paying American calls.	Both families produced coherent application messages.
Parameters	Plan controlled ranges for r , q , σ , and T .	Planning report and methodology handouts.
Solver identity	Use baseline CN/PSOR as the default source.	Ticket 12 and Ticket 10A.
Splits	Use regime-level train/test splits, not only random point splits.	Handout warns that surrogate credibility depends on held-out regimes.
Core quantities	Preserve value, payoff, continuation premium, and parameter metadata.	Solver and pilot reports.
Boundary metadata	Preserve threshold, interpolation, and found/not-found status.	Ticket 09 and analytics evidence.
Greek masks	Preserve kink, maturity, boundary, and strict-mask flags.	Ticket 10 and analytics Gamma caution.
Diagnostics	Require PSOR convergence and LCP residuals for every solved regime.	Ticket 08 and quantitative analytics.

The machine-readable policy companion is:

`reports/04_downstream/dataset_readiness_policy.csv`

That file is a planning policy table. It is not a dataset.

9 Human Review Checklist

Review question	Required answer before dataset construction
Is baseline CN/PSOR accepted as the initial label source?	Yes, or document a new method gate.
Are future parameter ranges small enough for a first dataset?	Yes, with explicit ranges and rationale.
Are held-out regimes defined before generation?	Yes.
Are boundary thresholds and interpolation rules fixed?	Yes, with metadata fields.
Are Gamma masks specified?	Yes, especially near $S = K$, maturity, and boundary bands.
Are PSOR and LCP diagnostics required for every generated surface?	Yes.
Are dataset files still blocked until the construction plan is approved?	Yes.
Are neural models still blocked until after dataset review?	Yes.

10 Limitations

- The application analytics are small and controlled; they are not broad stress-regime coverage.
- Boundary curves are useful but threshold-based.
- Gamma remains diagnostic, not production-quality.
- Runtime evidence is local to the current machine and grid size.
- The methodology handouts describe later dataset and surrogate stages, but this gate does not implement those stages.
- The decision allows design planning only; it does not approve generation.

11 Recommended Next Step

The recommended next step is a surrogate dataset design plan. That plan should define the future schema, parameter ranges, split strategy, output labels, mask fields, diagnostic acceptance criteria, and human-review checkpoint. It should also identify optional higher-grid confirmation cases before any large generation run.

The project should keep the current gate language visible: the application evidence is strong enough to plan the next stage, but not strong enough to construct datasets or train neural surrogates yet.

12 Reproducibility and Source Record

Item	Record
Ticket 12 synthesis	reports/03_solver/tickets/ticket_12_solver_validation_synthesis.tex.
Downstream gate	reports/04_downstream/downstream_design_gate.tex.
Pilot 01 report	reports/04_downstream/pilot_stress_maps_report.tex.
Quantitative analytics report	reports/04_downstream/quantitative_risk_analytics_report.tex.
Pilot tables	results/02_pilot_stress_maps/tables/.
Analytics tables	results/03_quantitative_risk_analytics/tables/.
Policy CSV	reports/04_downstream/dataset_readiness_policy.csv.
Bibliography	reports/03_solver/references.bib.

References

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